

Universal derivative-square model, Cameron–Martin action, and raw-jet obstruction

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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Purpose and scope

One exact stochastic-model subgate is closed, and an ambiguous coefficient-jet claim is corrected before it can become an overclaim. For the hash-pinned production Gaussian, this note constructs the universal centered derivative-square tensor in every finite probability moment and proves its deterministic H^2 Cameron–Martin translation continuity. The only divergent subtraction at this Q level is the derivative covariance.

The advertised regularities of the later XQ and XXQ coefficient jets are false if the diamond is read as a full Wick product. For a cone-localized piece of a natural unnormalised nested resonance, one scalar double pairing has a fixed negative coefficient whose magnitude stays positive on every dyadic scale. This is a contraction-classification certificate, not a proof that the total symmetric Littlewood–Paley tree has a nonzero logarithmic coefficient. Thus the parent translated-current model gate is not closed by silently choosing either reading. Its coefficient-jet half is split into an explicit renormalisation-classification gate. A13 remains T4, and no Nelson or interacting-measure theorem is claimed.

The scope is the fixed $L = 16$ three-torus, the six-real form of the A1 production Gaussian, the fixed positive density floor inherited by the Class-II coefficients, common real-even scalar spectral multipliers coupled through the same Gaussian modes, and exact or sufficiently de-aliased products. Admitted multipliers are uniformly bounded approximate identities that converge pointwise to one and use the same real-even scalar multiplier for every component. Cameron–Martin shifts are deterministic elements of H^2 . Adapted random shifts are outside this theorem.

1. The universal centered tensor

Let $\widehat{C}(k)$ denote the six-real covariance symbol. The production ellipticity bounds used by A6 and A7 give constants $0 < c < C < \infty$ such that at high momentum

$$c\langle k \rangle^{-4}\mathbf{I} \leq \widehat{C}(k) \leq C\langle k \rangle^{-4}\mathbf{I}. \quad (1.1)$$

Let R_Λ be an admitted common scalar multiplier and put $X_\Lambda = R_\Lambda X$. For a spatial direction i and real component indices a, b , define

$$\Gamma_{\Lambda,i}^{ab} = \mathbb{E}[\partial_i X_\Lambda^a(x) \partial_i X_\Lambda^b(x)], \quad Q_{\Lambda,i}^{ab} = \partial_i X_\Lambda^a \partial_i X_\Lambda^b - \Gamma_{\Lambda,i}^{ab}. \quad (1.2)$$

Theorem 1.1. For every $\kappa > 0$ and every finite p , there is a regulator-independent tensor Q_i^{ab} such that

$$Q_{\Lambda,i}^{ab} \longrightarrow Q_i^{ab} \quad \text{in} \quad L^p(\Omega; H^{-1-\kappa}(\mathbb{T}_L^3)). \quad (1.3)$$

The convergence is joint over the finite family of directions and component indices and with the A6 enhancement $(X, \mathbb{X})$, where $\mathbb{X}_i^{ab} = X^a \circ \partial_i X^b$. A subsequence selected from the dyadic cutoffs converges almost surely. The A6 area theorem is inherited rather than restated as a new result here.

1.1 Fourier–Wick estimate

Apart from harmless torus normalisations, the Fourier coefficient of (1.2) is

$$\widehat{Q}_{\Lambda,i}^{ab}(n) = - \sum_{k+\ell=n} k_i \ell_i : \widehat{X}_{\Lambda}^a(k) \widehat{X}_{\Lambda}^b(\ell) : . \quad (1.4)$$

Wick isometry, the finite internal dimension, and (1.1) imply

$$\mathbb{E}|\widehat{Q}_{\Lambda,i}^{ab}(n)|^2 \leq C \sum_{k+\ell=n} \langle k \rangle^{-2} \langle \ell \rangle^{-2}. \quad (1.5)$$

The discrete convolution bound in three dimensions is

$$\sum_k \langle k \rangle^{-2} \langle n-k \rangle^{-2} \leq C \langle n \rangle^{-1}. \quad (1.6)$$

Indeed, on $|k| \leq |n|/2$ the second factor is $O(\langle n \rangle^{-2})$ and the first sum is $O(\langle n \rangle)$; the region $|n-k| \leq |n|/2$ is symmetric. On the remaining region both factors are at most their common-scale bounds, and dyadic summation again gives $O(\langle n \rangle^{-1})$.

Consequently

$$\mathbb{E}\|Q_{\Lambda}\|_{H^{-1-\kappa}}^2 \leq C \sum_n \langle n \rangle^{-2-2\kappa} \langle n \rangle^{-1} < \infty. \quad (1.7)$$

The radial summand is $O(r^{-1-2\kappa})$, explaining both convergence for every positive κ and exclusion of the H^{-1} endpoint.

Couple two cutoffs through the same Gaussian modes. At every fixed Fourier configuration their multiplier difference tends to zero, while (1.5)–(1.7) give a summable cutoff-independent majorant. Dominated convergence proves the $L^2(H^{-1-\kappa})$ Cauchy property and regulator independence.

1.2 Every finite probability moment

Each $Q_{\Lambda} - Q_{\Lambda'}$ lies in the second homogeneous Wiener chaos. For a Hilbert-valued chaos of degree m and $p \geq 2$, Nelson hypercontractivity gives

$$\|Z\|_{L^p(\Omega;\mathcal{H})} \leq (p-1)^{m/2} \|Z\|_{L^2(\Omega;\mathcal{H})}. \quad (1.8)$$

Apply (1.8) with $m = 2$ and $\mathcal{H} = H^{-1-\kappa}$. For $p < 2$, probability monotonicity suffices. Thus (1.3) holds for every finite p , not merely for $p = 3$. This distinction is load-bearing: in the coefficient-model application range $0 < \kappa < 1/2$, the translation-model estimate asks for

$$p_{\kappa} = \frac{6}{2-\kappa} > 3. \quad (1.9)$$

At the audited $\kappa = 0.1$, $p_{\kappa} = 60/19$.

2. Cameron–Martin translation

For deterministic $h \in H^2(\mathbb{T}_L^3; \mathbb{R}^6)$ define

$$\begin{aligned} (\tau_h Q)_i^{ab} &= Q_i^{ab} + \partial_i h^a \partial_i X^b + \partial_i X^a \partial_i h^b \\ &\quad + \partial_i h^a \partial_i h^b. \end{aligned} \quad (2.1)$$

At every finite cutoff this is the exact algebraic identity obtained by replacing X_Λ with $X_\Lambda + h_\Lambda$ while keeping the same Gaussian covariance subtraction $\Gamma_{\Lambda,i}$.

Theorem 2.1. For every finite p and every $\kappa > 0$, (2.1) is an element of $L^p(\Omega; H^{-1-\kappa})$. On each H^2 ball,

$$\|\tau_h Q - \tau_g Q\|_{L^p(\Omega; H^{-1-\kappa})} \leq C_{p,\kappa,R}(1 + \|h\|_{H^2} + \|g\|_{H^2})\|h - g\|_{H^2}. \quad (2.2)$$

For each fixed h , if $h_\Lambda = R_\Lambda h$, then $\tau_{h_\Lambda} Q_\Lambda \rightarrow \tau_h Q$ in the same space.

Proof. It is enough first to take $0 < \bar{\kappa} = \min\{\kappa, 1/4\}$. For $R_h = \partial h \partial X$, Fourier orthogonality and (1.1) give

$$\begin{aligned} \mathbb{E}\|R_h\|_{H^{-1-\bar{\kappa}}}^2 &\leq C \sum_k |k|^2 |\hat{h}(k)|^2 \sum_m \langle m+k \rangle^{-2-2\bar{\kappa}} \langle m \rangle^{-2} \\ &\leq C_{\bar{\kappa}} \|h\|_{H^2}^2. \end{aligned} \quad (2.3)$$

The inner convolution is uniformly bounded because the two decay exponents sum to $4 + 2\bar{\kappa} > 3$. The random term is first chaos, so (1.8) gives every finite p . Moreover $H^1 \hookrightarrow L^4$ on the finite torus, and hence

$$\|\partial h \partial g\|_{L^2} \leq C \|h\|_{H^2} \|g\|_{H^2}. \quad (2.4)$$

Equations (2.3)–(2.4), applied to differences, prove (2.2). If $\kappa > 1/4$, embed the already constructed $H^{-1-1/4}$ object into $H^{-1-\kappa}$. Finally $R_\Lambda h \rightarrow h$ strongly in H^2 for each fixed h , and the coupled Gaussian cutoff convergence follows from the same dominated Fourier estimate. $\square$

This theorem is deterministic in h . Substituting an adapted random $h = h(X)$ requires joint moments and measurability estimates not supplied by (2.2).

3. Counterterm classification at the closed subgate

The value covariance is summable. On unit radial shells,

$$\sum_{r \leq |k| < r+1} \langle k \rangle^{-4} = O(r^{-2}), \quad (3.1)$$

whereas the derivative covariance has order one per unit radial shell:

$$\sum_{r \leq |k| < r+1} |k|^2 \langle k \rangle^{-4} = O(1). \quad (3.2)$$

Thus $\Gamma_{\Lambda,i}$ is the primitive divergent local contraction in Q . Common real-even multipliers also give the exact full same-point identity

$$\mathbb{E}[X_\Lambda^c(x) \partial_i X_\Lambda^a(x)] = \sum_k i k_i \hat{C}^{ca}(k) |R_\Lambda(k)|^2 = 0. \quad (3.3)$$

Translation by deterministic h introduces no new stochastic pairing. Hence $\Gamma_{\Lambda,i}$ is the only divergent Q -level local subtraction.

Equation (3.3) must not be overextended. A Littlewood–Paley resonance selector need not preserve the global $k \mapsto -k$ cancellation inside each localized contraction. Such lower-chaos terms belong to the next model classification, not to the closed Q theorem.

4. Why full coefficient products do not have the target gain

Put $D_i(k) = k_i^2 \hat{C}(k)$. On a fixed cone, ellipticity gives $D_i(k) \asymp |k|^{-2}$. Restricting the convolution to a ball of order $|n|$ around $n/2$ gives the matching lower bound

$$\mathbb{E}|\hat{Q}_i(n)|^2 \geq c|n|^{-1}. \quad (4.1)$$

The value covariance is in $\ell^1(\mathbb{Z}^3)$ and has a positive zero mode. Therefore the full Wick products retain the same low-high tail:

$$\mathbb{E}|\widehat{XQ}:(n)|^2 \geq c|n|^{-1}, \quad \mathbb{E}|\widehat{XXQ}:(n)|^2 \geq c|n|^{-1}. \quad (4.2)$$

A spectrum $|n|^{-1}$ in dimension three has finite $L^2(\Omega; H^s)$ model norm only for $s < -1$. For small $\kappa < \alpha$, this contradicts both advertised full-product targets

$$s = \alpha - 1 - \kappa > -1, \quad s = 2\alpha - 1 - \kappa > -1. \quad (4.3)$$

Thus the diamonds in the parent gate cannot mean literal full products. This is an analytic no-go, not a failure of the centered Q tensor.

5. Cone-localized raw nested-resonance certificate

It is also unsafe to assume that renaming the diamonds as unnormalised nested Bony products removes every lower-chaos contraction. Consider the natural tree

$$\Pi(X, \Pi(X, Q)). \quad (5.1)$$

For a dyadic integer r , let

$$S_r = \{k \in \mathbb{Z}^3 : r \leq k_j < 2r, \ j = 1, 2, 3\}. \quad (5.2)$$

Insert a broad balanced angular localization which is one on the following configurations. Choose the two value-field momenta $\ell, j \in S_r$ and pair the two derivative legs of Q with $-\ell, -j$. The inner Q output is $-(\ell + j)$, the inner product output is $-\ell$, and the outer product has zero output. All inputs are comparable in scale, so this is a legitimate cone-localized balanced-resonance subdiagram. No assertion about an unspecified global Littlewood–Paley multiplier is made.

For the scalar covariance $(1 + |k|^2)^{-2}$ used by the executable certificate, the magnitude of one perfect matching is

$$A_r = \left(\sum_{k \in S_r} k_1 (1 + |k|^2)^{-2} \right)^2. \quad (5.3)$$

Since $|S_r| = r^3$, $k_1 \geq r$, and $|k|^2 < 12r^2$,

$$A_r \geq \left(\frac{r^4}{(1 + 12r^2)^2} \right)^2 \longrightarrow \frac{1}{12^4} > 0. \quad (5.4)$$

With the sign convention in (1.4), the scalar double-cross coefficient from the two perfect matchings is $-2A_r$, not $+A_r$. Thus its absolute magnitude has a scale-independent positive lower bound. The production ellipticity (1.1) gives the same conclusion, with a changed constant, for a diagonal production component. Summing this cone-localized component over the $\asymp \log_2 \Lambda$ admitted scales gives magnitude at least $c \log \Lambda$.

This does *not* prove that the total symmetric Littlewood–Paley tree has the same logarithmic coefficient: other angular sectors may cancel it. It proves that global same-point evenness and superficial power counting do not classify the lower-chaos terms. The next gate must fix the actual multiplier and calculate the total coefficient, or use base-point/homogeneous-chaos correctors for which the cancellation is explicit.

The first raw resonance $\Pi(X, Q)$ similarly has two possible value-derivative contractions. A localized selector generally leaves a finite, nonzero first-chaos Fourier multiplier even though the sum in (3.3) vanishes. At second order there are four single cross contractions and two double cross contractions. A valid coefficient model must specify their homogeneous-chaos projections or equivalent correctors explicitly.

This does not prove that the Class-II action, or even the final specified total tree, needs a new physical counterterm. A7 proves convergence of one exact $B(X)$ -contracted scalar with the derivative covariance subtraction. The open task is to show that all model-level lower-chaos normalisations cancel under the translated coefficient reconstruction, or else to register the surviving term instead of hiding it.

6. Corrected proof order

The subgate

A13-CLASSII-UNIVERSAL-Q-ALL-MOMENTS-AND-CM-TRANSLATION

is closed by Theorems 1.1 and 2.1. The parent

A13-CLASSII-CAMERON-MARTIN-TRANSLATED-CURRENT-MODEL-LIFT

remains open. Its next subordinate gate is

A13-CLASSII-COEFFICIENT-JET-RENORMALISATION-CLASSIFICATION.

That gate must do all of the following.

1. Fix either base-point increment symbols or an exact balanced Littlewood–Paley corrector family for the first- and second-order jets.
2. Enumerate every first-, second-, and zero-chaos contraction, including both two- X /one- Q parenthesisations needed by second-order parilinearisation.
3. Prove coupled-cutoff convergence and every finite probability moment.
4. Reconstruct $\Delta B : Q$ and the translated current in the exact A7 covariance-normal scheme, without a finite Hermite truncation of the rational coefficient.
5. Prove cancellation of the logarithmic and finite lower-chaos model terms in that reconstruction, or register a surviving interaction counterterm as a new obstruction.

Only after that classification may the proof move to the strict-past signed Q /current/positive-square/Cartan-curvature form bound. Termwise absolute Young absorption remains excluded by the earlier A13 critical-exponent calculation.

7. Executable audit

The primary route derives every exponent from the dimension and covariance decay, forms the finite scalar Q convolution by zero-padded FFT, checks the exact translated tensor identity, and computes (5.3) on four dyadic cones. For $r = 2, 4, 8, 16$, the respective positive contraction magnitudes are approximately

$$0.00274457, \quad 0.00185506, \quad 0.00147026, \quad 0.00130040. \quad (7.1)$$

The localized first-chaos selector witness is -0.09512894227995705 , confirming that full parity does not kill every resonant contraction.

The non-importing route enumerates the Q convolution directly, checks the Wick translation polynomial with probabilists’ Hermite polynomials, verifies the degree-2, 3, 4 Nelson bounds by 120-node Gaussian quadrature, counts the cross contractions, and independently recomputes three cone lower bounds. These finite calculations are regression and falsifier evidence. The continuum theorem is (1.4)–(2.4), the full-product no-go is (4.1)–(4.3), and the cone-localized classification boundary is (5.1)–(5.4).

The one-command verifier is

```
python codes/foundations/a13_classii_universal_q_cm_translation_verify.py
```

Devil’s-advocate

1. **DISMISSED: A7’s L^2 result is enough because every translated coefficient is fixed chaos.** The rational $B(X)$ has infinitely many chaos components. Hypercontractivity is applied here only to the universal second-chaos tensor Q , not directly to A7’s nonlinear contraction.
2. **UPHELD AS A CORRECTION: exact L^3 is the required endpoint.** For the application range $0 < \kappa < 1/2$, (1.9) is strictly above three. The theorem therefore proves all finite moments.
3. **DISMISSED: $H^2 \hookrightarrow C^{1/2}$ permits an ordinary Holder product with ∂X .** That endpoint multiplication is not used. The mixed Fourier–Sobolev estimate (2.3) constructs the product directly.
4. **VALID WITH MITIGATION: local Lipschitz continuity for deterministic h proves the adapted case.** It does not. Joint moments for $h(X)$ and the signed stochastic form estimate remain open.
5. **UPHELD AS A NO-GO: the diamonds may be literal full Wick products.** Equations (4.1)–(4.3) show that their low–high spectra miss the advertised regularities.
6. **UPHELD AS A NO-GO: changing “full” to “resonant” removes all contractions by evenness.** The explicit cone (5.2)–(5.4) gives a fixed-sign zero-chaos subdiagram with logarithmically growing magnitude, and the first-chaos localized witness is nonzero. Total-tree noncancellation is not claimed.
7. **VALID WITH MITIGATION: the logarithm proves a new physical Class-II counterterm is necessary.** It proves only that a cone-localized subdiagram is not absolutely summable and must be included in the contraction classification. Whether it cancels across the specified total tree and the exact translated Class-II reconstruction is precisely the next gate.
8. **DISMISSED: the finite cone and FFT fixtures are presented as a continuum proof.** They are only independent regression certificates. The analytic majorants and lower bound are written explicitly in the note.

Result footer

Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION
Result: A13-CLASSII-UNIVERSAL-Q-ALL-MOMENTS-AND-CM-TRANSLATION
Tier: T4 (unchanged)
Closed: universal centered Q in every finite $L_p(H^{-(1-\kappa)})$; deterministic
H2 Cameron-Martin action and local Lipschitz continuity; Q -level
derivative-covariance counterterm classification.
Negative: full-product coefficient jets miss the targets; a cone-localized
nested QXX subdiagram has a fixed negative coefficient with
logarithmically growing magnitude. Total-tree noncancellation is
not claimed.
Open: A13-CLASSII-CAMERON-MARTIN-TRANSLATED-CURRENT-MODEL-LIFT and the
umbrella A13-CLASSII-CONTROLLED-SHELL-ENERGY-ONE-USE theorem.
Next: A13-CLASSII-COEFFICIENT-JET-RENORMALISATION-CLASSIFICATION.
Honesty: no adapted-shift form bound, Nelson moment, interacting measure,
floor/regulator removal, infinite volume, phase transition, BCC,
T5, T6, or T7 is proved here.